Макет №1

1. $\int \frac{x-1}{x^2-x-1}dx = \frac{1}{2}\int \frac{2x-1}{x^2-x-1}dx - \frac{1}{2}\int \frac{dx}{x^2-x-1} = \frac{1}{2}\int \frac{d(x^2-x-1)}{x^2-x-1} - \frac{1}{2}\int \frac{d(x-0{,}5)}{(x-0{,}5)^2-1{,}25} =$

$= \frac{\ln|x^2-x-1|}{2} - \frac{1}{5}\ln\left|\frac{x-1{,}75}{x+0{,}75}\right| + C$

2. $\int \frac{3x-6}{\sqrt{x^2-4x+5}}dx = \frac{3}{2}\int \frac{2x-4}{\sqrt{x^2-4x+5}}dx = \frac{3}{2}\int \frac{d(x^2-4x+5)}{\sqrt{x^2-4x+5}} = 3\sqrt{x^2-4x+5} + C$

3. $\int \frac{x^2-1}{x^4+1}dx = \int \frac{(1-\frac{1}{x^2})dx}{(x^2+\frac{1}{x^2}+1)-1} = \int \frac{d(x+\frac{1}{x})}{(x+\frac{1}{x})^2-1} = \frac{1}{2}\ln\left|\frac{x+\frac{1}{x}-1}{x+\frac{1}{x}+1}\right| + C \ominus$

$\ominus \frac{1}{2}\ln\left|\frac{x^2-x+1}{x^2+x+1}\right| + C$

4. $\int \frac{e^x dx}{\sqrt{4-e^{2x}}} = \int \frac{d e^x}{\sqrt{2^2-(e^x)^2}} = \arcsin\frac{e^x}{2} + C$

5. $\int \frac{dx}{x\ln x \ln\ln x} = \int \frac{d\ln x}{\ln x \ln\ln x} = \int \frac{d\ln\ln x}{\ln\ln x} = \ln\ln\ln x + C$

6. $\int \frac{\sin x}{\sqrt{1+2\cos x}}dx = -\frac{1}{2}\int \frac{d(2\cos x+1)}{\sqrt{1+2\cos x}} = -\sqrt{1+2\cos x} + C$

7. $\int \frac{\ln\operatorname{tg} x}{\sin 2x}dx = \frac{1}{2}\int \frac{\ln\operatorname{tg} x}{\operatorname{tg} x}d\operatorname{tg} x = \frac{\ln^2\operatorname{tg} x}{4} + C$

8. $\int \frac{\operatorname{arctg}\sqrt{x}}{(1+x)\sqrt{x}}dx = 2\int \frac{\operatorname{arctg}\sqrt{x}}{1+(\sqrt{x})^2}d\sqrt{x} = \operatorname{arctg}^2\sqrt{x} + C$

9. $\int \frac{\operatorname{arccos}^2 2x}{\sqrt{1-4x^2}}dx = -\frac{1}{2}\int \operatorname{arccos}^2 2x\, d\arccos 2x = -\frac{\operatorname{arccos}^3 2x}{6} + C$

10. $\int \operatorname{arccos}(5x-2)dx = \frac{1}{5}\int \operatorname{arccos}(5x-2)d(5x-2) = \frac{(5x-2)\operatorname{arccos}(5x-2)}{5} - \frac{\sqrt{1-(5x-2)^2}}{5} + C$

$\triangle \int \operatorname{arccos} x\,dx = \left|\begin{matrix} u=\arccos x,\ du=-\frac{dx}{\sqrt{1-x^2}} \\ dv=dx,\ v=x \end{matrix}\right| = x\arccos x + \int \frac{x\,dx}{\sqrt{1-x^2}} =$

$= x\arccos x - \frac{1}{2}\int \frac{d(1-x^2)}{\sqrt{1-x^2}} = x\arccos x - \sqrt{1-x^2} + C$

11. $\int \frac{\arcsin x}{x^2}dx = \left|du = \frac{dx}{1-x^2},\ v = -\frac{1}{x}\right| = -\frac{\arcsin x}{x} + \int \frac{dx}{x(1-x^2)} = -\frac{\arcsin x}{x} +$

$-\frac{1}{2}\int \frac{d\frac{1}{x^2}}{\frac{1}{x^2}-1} = -\frac{\arcsin x}{x} - \frac{1}{4}\ln\left|\frac{\frac{1}{x}-1}{\frac{1}{x}+1}\right| + C = -\frac{\arcsin x}{x} - \frac{1}{4}\ln\left|\frac{1-x}{1+x}\right| + C$

12. $\int \left(\frac{\ln x}{x}\right)^3 dx = \left|\begin{matrix} u=\ln^3 x,\ du=3\frac{\ln^2 x}{x}dx \\ dv=\frac{dx}{x^3},\ v=-\frac{1}{x^2} \end{matrix}\right| = -\frac{\ln^3 x}{x^2} + 3\int \frac{\ln^2 x}{x^3}dx = \left|\begin{matrix} u=\ln^2 x,\ du=\frac{2\ln x\,dx}{x} \\ v=-\frac{1}{x^2} \end{matrix}\right| =$

$= -\frac{\ln^3 x}{x^2} - \frac{3\ln^2 x}{x^2} + 6\int \frac{\ln x\,dx}{x} = -\frac{\ln^3 x}{x^2} - \frac{3\ln^2 x}{x^2} + 3\ln^2 x + C$

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14. $\int x e^x \sin x\,dx = \left| \begin{matrix} u = x, & du = dx \\ dv = e^x \sin x dx, & v = \frac{-e^x \cos x + e^x \sin x}{2} \end{matrix} \right| =$

$\Delta \int e^x \sin x\,dx = \left| \begin{matrix} u = e^x, & du = e^x dx \\ dv = \sin x dx, & v = -\cos x \end{matrix} \right| = -e^x \cos x + \int e^x \cos x\,dx =$

$\Delta \int e^x \cos x\,dx = \left| \begin{matrix} u = e^x, & du = e^x dx \\ dv = \cos x dx, & v = \sin x \end{matrix} \right| = e^x \sin x - \int e^x \sin x\,dx$

$= -e^x \cos x + e^x \sin x - \int e^x \sin x\,dx \Rightarrow \int e^x \sin x\,dx = \frac{-e^x \cos x + e^x \sin x}{2}$

$= \frac{-x e^x \cos x + x e^x \sin x}{2} + \frac{1}{2}\int e^x \cos x\,dx - \frac{1}{2}\int e^x \sin x\,dx =$

$= \frac{x e^x \sin x - x e^x \cos x}{2} + \frac{e^x \sin x - e^x \cos x}{2} - \frac{e^x \sin x - e^x \cos x}{2} =$

$= \frac{e^x}{2}(x \sin x + (1-x)\cos x) + C$

15. $\int \cos^5 x\,dx = \int (1 - \sin^2 x)^2\, d \sin x = \int (1 - 2\sin^2 x + \sin^4 x)\, d \sin x = \sin x - \frac{2\sin^3 x}{3} + \frac{\sin^5 x}{5} + C$

16. $\int \frac{x^2 e^x}{(x+2)^2}\,dx = \int \left(1 - \frac{2}{x+2}\right)^2 e^x dx = \int \left(1 - \frac{4}{x+2} + \frac{4}{(x+2)^2}\right) e^x dx =$

$= e^x - 4\int \frac{e^x dx}{x+2} + 4\int \frac{e^x dx}{(x+2)^2} = \left| \begin{matrix} u = \frac{1}{x+2}, & du = -\frac{1}{(x+2)^2} \\ dv = e^x dx, & v = e^x \end{matrix} \right| = e^x - 4\frac{e^x}{x+2} - 4\int \frac{e^x dx}{(x+2)^2} + 4\int \frac{e^x dx}{(x+2)^2} =$

$= e^x\left(1 - \frac{4}{x+2}\right) + C$

13. $\int x^2 (x^2+1)^{\frac{1}{2}}\,dx = \left| t = 1 + \frac{1}{x^2} \right| = \int \frac{t^2}{(t^2-1)^3}\,dt = \ldots$ (раскладываем на простейшие, находим коэффициенты, интегрируем по Ln и всё)